Review: Power System Analysis Software Tools

Lokendra Bam, *Student Member, IEEE*, and Ward Jewell, *Fellow, IEEE*

Abstract— **This paper summarizes the various user-friendly options that power system analysis and simulation software tools offer today with varied add-ons and premium modules. The authors' main intention is to present the information and key features of available software tools in power engineering. Based on the study presented in this paper, one could conclude that the advanced analysis methodology combines generation, transmission, various elements, and market economics into one virtual environment to deliver location, time, user-dependent analysis results, and indicators. The authors hope that the information furnished in this paper is helpful to potential users in industry and academia, and enables them to make the right decision before purchasing appropriate software.**

Index Terms—**Power System, Analysis, Simulation Software, Software Packages, Software Tools, Computer Applications.**

I. Nomenclature

AGC	Automatic Generation Control
ASPEN	Advanced Systems Control
ATP	Alternate Transient Program
CAI	Commonwealth Program
CAPE	Computer-Aided Protection Engineering
DIgSILENT	DIgital SImuLatro for Electrical NeTworks
DSA	Dynamic Security Assessment
DSTAR	Distribution System Testing Application and Research
EMTP	Electromagnetic Transient Program
GE PSLF	General Electric Positive Sequence Load Flow
MAPS	Multi-Area Production Simulation Software
MARS	Multi-Area Reliability Simulation Software
MUST	Managing and Utilizing System Transmission
PSAF	Power System Analysis Framework
PSCAD	Power System Computer Aided Design
PSDS	Power System Dynamic Simulation
PSMS	PowerStation Management System
PSS	Power System Simulation
PSS/ADEPT	…Advanced Distribution Engineering Productivity Tool
PSS/E	…Engineering
PSS/O	…Operation
PSS/VIPER	…VIsual Power EngineeRing
RTDS	Real Time Digital Simulators
SCSC	Short Circuit Calcutations
SPICE	Simulation Programs for Integrated Circuits
SPS	SimPowerSystem
STAR	System Protection & Device Coordination
TACS	Transient Analysis of Control Systems
TPLAN	Transmission Reliability Assessment

II. Introduction

The electric industry serves one of the most rapidly changing and enormously demanding needs of the nation. As predicted by experts, capital for expansion of the electric infrastructure in addition to capital for operation within the set constraints (efficiency, environment, and economy) are vital to the sustenance of utilities and electric companies.

The fundamental need of a utility or power company is two-fold: efficient planning and accurate prediction of equipment life with an emphasis on avoiding of over-use or under-use of the resources available. Modeling power system requirements for generation, transmission and distribution of electric power is needed to make real-value decisions regarding capital investment and asset management. Factors associated with modeling of power systems using an efficient analytical tool include the following:

- Increased productivity
- Reduced capital expenditure
- Reduction in obsolete parts inventory
- Reduction in repairs
- Return on investment after first year of implementation
- Improved services

Constrained probabilistic models are used to predict the remaining life of equipment. Probabilistic modeling provides an index of the frequency of faults on the equipment under study. Fault-studies involving varying amplitudes of the fault-causing variables could be effectively simulated with power system analysis tools. With the information from probabilistic models and power system analysis tools, the remaining life of equipment can be determined on a time-domain reference. This would eventually help planning personnel to make the correct decisions regarding expansion and system maintenance.

The power system analysis software tools mentioned in this paper are not limited to their key technical features but also include individual capabilities for system modeling, design and analysis. Some main features of a tool include:

- ✓ Power Flow
- ✓ Short Circuit
- ✓ Stability
- ✓ Protection and Coordination
- ✓ Voltage Control/AGC
- ✓ DC System Analysis

0-7803-9156-X/05/$20.00 ©2005 IEEE.

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.

- ✓Optimal Power Flow
- ✓Contingency
- ✓Reliability
- ✓Availability of Component Models
- ✓Integration/Database/Reporting Capabilities
- ✓Power Market Modeling and Analysis

III. Evaluation methodologies

This section will summarize features of a power system analysis software tool (PSA ST). A number of tools are developed and utilized independently in the analysis and design of power systems. Power systems are classed as follows: transmission, distribution, and industrial [1]. Since a tool is developed for a specific class of the system in order to improve analytical ability as well as computational efficiency, its associated scope and limitations must be observed.

Proper modeling of high-voltage interconnecting transmission lines, regulating transformers, and generating units is most important for the protection and integration of the power system during normal and transient modes. For a radial low-voltage distribution system, maintaining the phase balance, power factor, and few-cycle duty protection are significant features to consider when selecting an analytical tool. However, most of the industrial systems, composed of radial as well as loop type having cogeneration capabilities, are serving to more voltage dependent and nonlinear electronics loads and equipment. Hence, depending on the nature of the modeling system and purpose of the analysis of a class or equipment, PSA STs for basic power flow, short-circuit analysis to transients, and harmonic analysis are currently available and in increasing demand. In this restructured and deregulated system, the models of power market and economic play vital roles in modeling and simulation. The following features may help users to evaluate a PSA ST before contracting or purchasing the product:

A. Basics behind key features

Some of the most basic analysis tool features provided by a PSA ST are load flow analysis, short-circuit analysis, motor analysis, voltage regulations, DC systems and/or devices analysis [2]. Behind each analysis study is mathematics. An awareness of mathematical analysis is important to obtain fast, efficient, and reliable results. Gauss-Seidel (GS), Newton-Raphson (NR), and Fast-Decoupled (FD) iterative numerical methods, for example, are most commonly used to solve the power flow equations [3]. Combined algorithms of GS and NR can be successfully utilized to achieve a high rate of convergence and less sensitivity to initialization. Various studies show that NR is suited for distribution systems, while FD is suited for transmission systems [1]. Nonlinear loads are represented as current or voltage harmonics sources whose characteristics are determined using Fourier expansion analysis (FEA) of the load.

B. Mathematics behind models [solver technology]

The modeling structure will vary, depending on the nature of the actual physical or hypothetical system and the purpose of the simulation. Basically, simulation is a technique that involves setting up a model of a real situation and performing experiments on the model. Therefore, the models developed in a PSA ST are characterized by a mix of differential and algebraic equations [4]. Mathematical models may be classified as follows:

- Linear or Nonlinear
- Lumped or Distributed Parameter
- Static and Dynamic
- Continuous or Discrete
- Deterministic or Stochastic

Hence besides the simplifications and assumptions, a PSA ST has to be followed for the above models.

C. Simulation modes

There are some simulation modes applied in evaluation of a model design in the particular software tool. In addition to widely used continuous (steady state or transient) software, dynamics simulation software, instantaneous value simulation software, discretization simulation and phasor simulation software are also available.

For example, power electronic devices need to be discritized with fixed-time-step trapezoidal integration. The discretization mode also helps to simulate the desired model in real-time (dynamic simulation) whereas phasor simulation represents a set algebraic equation at a fixed frequency which facilitates the execution of transient stability studies of multi-machines design.

D. Analysis methods

An analysis of a system begins by defining initial states and conditions and using proper solver techniques to simulate the design model. Some analysis methods are as follows [5]:

- Steady-State Analysis: assumes sinusoidal voltages and currents, and linear system components; used to determine power flow; voltage profiles, losses (active and reactive), reactive power compensation, and transformer tap positions.
- Quasi-Steady State Analysis: used for abnormal system conditions, such as short circuit and harmonics.
- Dynamic Analysis: used to verify that the power system will not become unstable or even collapse during major disturbances, and to determine the operating limits of the system.
- Transient Analysis: simulate (with programs like EMTP) real-life power systems, including nonlinear models; unbalance impedances, and frequency - dependent parameters.

E. Multidomain environment

Model verifications are generally tested in either Time

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.

Domain (TD) and/or Frequency Domain (FD). A , model's accuracy can be determined by observing the responses received from simulation in TD and FD environments.

F. Components and blocks library facilities

The product should contain various electrical and mechanical components. Basic blocks include electrical sources, machines models, connectors, circuit elements and control and measurement blocks. Single-phase and three- phase components make the robust design of models.

G. User-friendly and machine-system compatible

A most important feature for a PSA ST is its user-interface mode or graphical user interface (GUI). Some of the basic user-friendly features include drag and drop style of component selection, design facilities, screen-editing facilities, display and reporting features, multi-color, and multi-dimensions presentation. The product should also be platform and machine independent.

Other important features include capability of import/export of data, multilevel undo/redo, and help utility and third-party product compatible. Since features may not be included in the primary product, the package should be capable of future "add-ons."

H. Industry standards

The product's design, components, and modules should be developed according to established industry standards, such as IEEE, NEC (NFPA), NESC, IEC. Various standards have been established for different individuals and groups.

I. Purchasing policy

A software product is a continuous development process. Every year, present new version that must be purchased in order to run a system smoothly without disturbing quality of service and product. Therefore, care must be taken while buying or installing a PSA ST for the system analysis and design. The main factors taken into considerations before the purchasing of software are availability of evaluation copy, product support, upgrading facility, price for upgrade and add-on tools, and obviously how far the product is useful,

IV. Products, developers and vendors

This section presents a brief overview of PSA ST products that are available in the market. Product details can be found on their respective home pages mentioned in the reference section.

i. PowerWorld Simulator [6]

PowerWorld is a comprehensive, robust power flow solution engine capable of efficiently solving systems of up to 60,000 buses. Basically, this power system analysis package is designed to simulate high voltage power system operation on a time frame ranging from several minutes to several days. The package is "extremely user-friendly and highly interactive", including one-line diagrams with animated flow, color contours, support for panning, zooming, and conditional display of all or a subset of various devices available in the system, such as buses, loads, generators to areas, and zones. This package is not only limited to transmission line simulation but also simulates OPF, security-constrained OPF, and hence is useful for simulating power market and internal education of power systems

ii. SimPowerSystem [7]

SimPowerSystem of TransÉnergie Technologies, Inc. of Montreal and marketed by The MathWorks, Inc. models and simulates electrical power systems within the Simulink/Matlab (multi-domain) environment. It has powerful capabilities for modeling the generation, transmission, and distribution of electrical power, especially when designing associated control systems. It contains more than 130 blocks for components and devices commonly found in electrical power networks, based on well-known electromagnetic and electromechanical equations, and are distributed in six libraries including sources, elements, machines, electronics, control and measurements, and three-phase components. The package can be used to develop for general purpose, user-defined modules, and user can also add his/her own modules in equation form.

iii. PSCAD [8]

PSCAD of Manitoba HVDC Research Centre is a fast, accurate and easy-to-use power system simulator for the design and verification of all types of power systems. It is a powerful and flexible design tool and graphical user interface to EMTDC. It comes with a library of pre-programmed and tested models, ranging from simple passive elements and control systems to more complex models, such as machines, FACTS devices, lines, and cables.

iv. WindMil [9]

MilSoft's WindMil is designed to operate with maximum efficiency in the modeling and analysis of radial and loop electrical systems. WindMil is especially targeted to today's utility engineering and operation staffs in their design and analysis of distribution systems by feeder, substation or entire system. It has a graphical display, editing tools, and a single-line diagram with GIS facility embedded with a LandBase tool. It also has a database of protective devices, LightTable, for protection coordination in the system.

v. PSS/E [10]

PSS/E of Shaw Power Technology, Inc., is an integrated program for simulating, analyzing, and optimizing power system performance. It has a one-line diagram with advanced features, such as animated flows and line-loading graphs. Several utilities and organizations use it for their power system analysis, and as a planning and marketing tool. Other add-on stand-alone application tools developed by PTI include PSS/ADEPT for distribution planning, PSS/Engines for various engineering

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.

calculations, PSS/VIPER for off-line power system analysis tool, MUST, a PC-base stand-alone product for managing and utilizing system transmission, and TPLAN for transmission reliability assessment

vi. GE PSLF [11]

GE power system analysis programs that can handle system models of up to 60,000 buses are structured as relational databases. GE's power system analysis package includes PSLF, SCSC, and PSDS programs for load-flow, short-circuit, and transient-stability work. Some of the key features of the package are load flow with relational database and graphics, dynamic simulation with plotting (PSDS), and a large-scale short-circuit calculations (SCSC). Several models of electrical elements are also available in the package. EPCL, a programming language, facilitates user-defined program, models that handle a wide range of system analysis and reporting tasks. Other stand-alone software include DSTAR Tools dedicated to routine distribution engineering task, GE MAPS dedicated to an hour-by-hour electric power market analysis and opportunities and production cost comparison depending on various system constraints, and GE MARS dedicated to system reliability assessment comprising a number of interconnected areas.

vii. PSAF [12]

CYME International's PSAF is integrated power system analysis software, equipped with simulation modules for conducting a wide array of system studies for utilities and industries. Currently available analysis modules of PSAF include CYMFLOW for power flow, motor starting, and AC contingency analysis, CYMFAULT for short-circuit analysis, CYMHARMO for harmonic analysis, CYMSTAB for transient-stability analysis, CYMLINE for one-line diagram, WECS for wind energy conversion systems, and CYME programs for line and cable parameter calculations. CYMDIST is another analysis package for distribution network planning and studies with various optional add-on modules.

viii. RTDS Simulator [13]

RTDS Technologies Inc. provides power systems simulation technology for real-time digital simulation study of power systems with complex high voltage alternating current (HVAC) and high voltage direct current (HVDC). The RTDS simulator with graphical user interface (GUI), RSCAD, software can be used for setting up simulations, controlling, and modifying power system parameters during a simulation, acquiring data, and analyzing results. Basically, the RTDS Simulator has two main software elements GUI (RSCAD) and a library of modules and compilers.

ix. ETAP [14]

ETAP of Operational Technology, Inc. is for the design, simulation, and analysis of generation, transmission, distribution, and industrial power systems. The ETAP software suite has various program modules like power system analysis, real-time simulation, advanced monitoring, optimization control, intelligent load shedding, energy usage cost analysis, and device coordination. The ETAP PowerStation family of programs is a fully integrated analysis tool used to design, maintain, and operate electric power systems in a totally graphical and virtual reality environment. Other integrated tools include ETAP PSMS for PowerStation Management System, ETAP STAR for System Protection and Device Coordination, and ETAP PanelSystems for panel system design and analysis.

x. ASPEN [15]

ASPEN is basically a stand-alone PC-based graphical user-machine interface for power system applications and analysis. It is equipped with integrated modules like Power Flow, DistriView, OneLiner, Breaker Rating Module, Relay Database, and Line Constants. The Power Flow program is designed for the planning, design and operating studies of transmission, sub-transmission and distribution networks. OneLiner is a short-circuit and relay coordination program for relay engineers. Relay Database is designed to be a repository of relay information for electric utilities and industrial facilities. DistriView is an integrated suite for voltage drop, short-circuit, and relay coordination software for utility distribution systems. The ASPEN modules also have data import-export facilities and built-in scripting language using the BASIC programming.

xi. ATP-EMTP [16]

ATP-EMTP is used world-wide for switching and lightning surge analysis, insulation coordination, shaft torsional oscillation studies, protective relay modeling, harmonic and power quality studies, and HVDC and FACTS modeling. ATP has been continuously developed through the international contributions of Drs. W. Meyer and Tsu-huei Lui. ATP development includes honesty in all dealings and non-participation in EMTP commerce, which is in the public domain. ATP is not in the public domain, and licensing is required before ATP materials are received. Details about this product and licensing can be found in the references [16].

xii. SPICE [17]

SPICE is a powerful general-purpose analog circuit simulator that is used to verify circuit designs and predict the circuit behavior. Especially for integrated circuits, SPICE was originally developed at the Electronics Research Laboratory of the University of California, Berkeley (1975). PSpice is a PC version of SPICE developed by MicroSim Corp (OrCAD), and HSpice is a version, developed by Avant! (Synopsys Corp.) that runs on workstations and larger computers. SPICE can perform many kinds of electronics circuits-level analysis, such as DC-AC, transient, noise, sensitivity, distortion, Fourier, and Monte Carlo. SPICE has an analog as well as digital library of all standard components.

xiii. EasyPower [18]

ESA's EasyPower is a suite of power system analysis software products that can be used for power system analysis, design, measurement, and control and that can be

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.

used to simulate a power system with unlimited busses. It offers an integrated one-line diagram with multiple analysis functionalities. The EasyPower suite of products includes EasyPower ShortCircuit, EasyPower PowerFlow, EasyPower PowerProtector (for protective device coordination and device database), EasyPower Spectrum (for harmonic and PQ problems), EasyPower ArcFlash, and EasyPower Measure.

xiv. PowerFactory [19]

DIgSILENT PowerFactory software is an integrated power system analysis tool that combines reliable and flexible system modeling capabilities with state-of-the-art solutions algorithms and unique object-oriented database management. One can develop detailed power system models in a single database with a wide range of steady-state, frequency-domain, time-domain, and stochastic system characteristics. The package supports the simulation and analysis for a system with a maximum of 100 buses, 250 buses, and an unlimited number of buses. Among the packages are PowerFactory Distribution, PowerFactory Transmission, PowerFactory Industrial, and PowerFactory Enterprise. PowerFactory is not only limited to analysis of respective sectors but also does reliability analysis, production planning, contingency analysis, power electronics device modeling, and interface for SCADA/GIS, and is compatible with a third-party product such as PSS/E & PSS/U.

xv. NEPLAN [20]

NEPLAN Electricity is user-friendly and fully integrated power system analysis software for transmission, distribution, and industrial networks, including OPF, transient stability, and reliability analysis. This program is developed by BCP Inc. in corporation with ABB Utilities GmbH and Swiss Federal Institute of Technology. NEPLAN Electricity is used to analyze, plan, optimize, and manage power networks for all levels of industrial and consumer supply voltages with any desire number of nodes.

xvi. SCOPE [21]

Nexant's SCOPE is a suite of software products that covers a wide range of power-flow-related simulation and optimization functions in electric power system operations control and planning, and today's deregulated electric energy markets. It also contains power flow, contingency analysis, and optimal power flow software. However, the key areas of SCOPE utilization include power engineering studies, online operation and control, and electric power market analysis.

xvii. CAPE [22]

CAPE, developed by Electrocon International, Inc., is the world's most powerful protection engineering software. It has an extensive database for protection engineering and the largest library of detailed relay models. CAPE consists of series of core and optional modules for analysis and reporting, linked by a general-purpose database. All CAPE modules contain an important set of standard features. CAPE's working space is user-friendly, GUI-based, which has the ability to define sets of buses, lines, transformers, and protective devices for batch operations or for use in macros (short programs). CAPE's user programming language (CUPL) helps to extend any module to perform an almost limitless number of useful, repetitive tasks such as CUPL macros that can be easily developed to perform automatic fault studies. The package includes built-in macros to compute fault locations, relay settings, reaches of instantaneous overcurrent and distance elements, generator breakers flashover currents, and state simulation data for end-to-end relay testing.

xviii. EDSA [23]

EDSA is a diverse company that not only markets numerous power system modeling tools but also is active in defense programs, nuclear power, electric propulsion, and advanced power system control [2]. EDSA's Technical 2000 program is the industry standard electrical design tool that helps a power system design a professional model, and simulate, manage, and market their electrical distribution and transmission designs. The Technical 2000 program uses state-of-the-art algorithms and object-oriented programming technology to enable engineers to conduct balanced, three-phase, and single-phase system load flow studies on almost any network configuration. The Technical 2000 with additional modules provides many capabilities for power system analysis, such as designing and simulating any electrical power system (T & D, Wind Farms, AC/DC networks), analyzing a power system from a variety of perspectives (steady-state, PQ, system reliability, dynamic analysis), import/export transferring, and being 100 percent AutoCAD compatible. Other features are available, depending on the package that is purchased.

xix. QuickStab Professional [24]

ECI's QuickStab Professional is a family of steady-state stability applications for power system operations and planning. The program uses the field-proven MultiArea QuickStab computational engine, enhanced with QuickStab Express, and comes with its own text editor all integrated with user-friendly GUI. It can be implemented real-time in multiple SCADA/EMS installations, and modular and simple data interfaced with third party applications.

xx. SPARD® mp [25]

Energy Computer Systems (ECS) SPARD® mp is affordable, operational, sophisticated, and easy to use software, which comprises the SPARD® mp Distribution package, SPARD® mp Power (power system simulator) and SPARD® mp Service Reliability Tracking. The Distribution package is used for operational analysis, optimization, and planning for distribution systems. The number of feeders, network elements, and buses are not theoretically limited and it is a total integration of a graphical/geographical user interface, applications, and database. The network management module makes available some features such as interface with EMS, network mapping and plotting, project costing and budgeting. SPARD® mp Power is also a GUI-based high speed power-flow simulator with complete models of most of the electrical components database.

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.

xxi. DSA PowerTools [26]

DSA PowerTools Suite, developed by Powertech Labs Inc., is a leading-edge software package for dynamic security assessment. The software is suitable for both off-line (system planning) and on-line (near-real-time) power system security assessment. DSA PowerTools include PowerFlow and Short Circuit Analysis Tool (PSAT), Voltage Security Assessment Tool (VSAT), Transient Security Assessment Tool (TSAT), Small Signal Assessment Tool (SSAT), and Control Design Tool (CDT).

xxii. TRANSMISSION 2000 [27]

CAI's TRANSMISSION 2000 is a highly interactive, powerful software product designed to improve the analysis and planning of electric utility transmission systems. TRANSMISSION 2000 comprises various modules for complete power system analysis including Power Flow (PFLOW), Constrained Economic Dispatch (CED), Contingency Processor (CP), Power Network Data Manager (PNDM), Power Flow Reduction and Data Conversion (PFR), Transient Stability (TS), Short Circuit (SC), Protection Analysis (PA), One-Line Diagram (OL), Line Constants (LC), Transmission Mapping for Network Analysis, and Transmission Access Information Library and PowerPlot.

xxiii. ABB Advise^{IT} GridView [28]

ABB Advise^{IT} GridView is a powerful energy market simulation and analysis tool design used to deal with the most challenging issues facing decision makers in the electric industries. In GridView, advanced analysis methodology combines generation, transmission, loads, fuels, and market economics into one integrated framework to deliver location-dependent market indicators, transmission system utilization measures, power system reliability, and market performance indices. It produces unit commitments and economic dispatches that respect the physical laws of power flow and transmission reliability requirements.

xxiv. Additional Tools

Each utility, power industry, and company has been using its own developed and customized PSA ST. However, the basic principles and purpose of all of them are common. Power* Tools SKM Systems Analysis, Inc. [29], SynerGEE Electric of Advantica Stoner [30], MicroTran of MicroTran Power System Analysis Corp.[31], and Interactive Power System Analysis (IPSA) software of IPSA Power Limited [32] are some other power system analysis-related software.

V. References

[1] N. N. Bengiamin, and F. H. Holcomb, "PC-Based Power Systems Software: Comparing Functions and Features," *IEEE Trans. Computer Applications in Power*, vol. 5, no. 1, pp. 35-40, Jan. 1992.

[2] D. T. Rizy, and R. H. Staunton "Evaluation of Distribution Analysis Software for DER Applications" ORNL/TM-2001/215, Sep. 30, 2002.

[3] J. D. Glover, and M. Sarma, *Power System Analysis and Design,* 2nd edition, PWS Publishing Company, Boston, MA, p. 253.

[4] Chee-Mun Ong, *Dynamic Simulation of Electric Machinery using MATLAB / SIMULINK,* 1998 by Prentice Hall PTR.

[5] Ljubomir Kojovic, "Modeling Requirements to Study Interactions of DG and Distribution Systems", *in proc. 2002, IEEE-PES Summer Meeting,*

[6] PowerWorld Corporation, *www.powerworld.com/features.html*

[7] The MathWorks, Inc., www.mathworks.com/products/simpower/, TransÉnergie Technologies,www.transenergie-tech.com/en/produits/simpowersystem.html

[8] Manitoba HVDC Research Center,*„ www.pscad.com*

[9] Milsoft Utility Solutions*, www.milsoft.com/milsoft/web.nsf/pages/engineering_analysis_overview/*

[10] Shaw Power Technologies, Inc. (Shaw PTI), *www.shawgrp.com/PTI/software/index.cfm*

[11] GE Energy, *www.gepower.com/prod_serv/products/utility_software /en/index.htm*

[12] CYME International, Inc., http://cyme.cognicase-hosting.com/products/.

[13] RTDS Technologies, Inc., *www.rtds.com*

[14] Operation Technologies, Inc. (OTI), *www.etap.com*

[15] Advanced Systems for Power Engineering, Inc., *www.aspeninc.com*

[16] W. Scott Meyer & Tsu-huei Liu, BPA*, www.emtp.org*

[17] MicroSim Corp. (OrCAD), *www.orcad.com /products/pspice/*;Avant! (Synopsys Corp.), www.synopsys.com; www.seas.upenn.edu/~jan/spice/spice.overview.html

[18] ESA Inc., www.easypower.com/easypower.htm

[19] DIgSILENT GmbH, www.digsilent.com/Software/PowerFactory_Features/

[20] Busarello+Cott+Partner Inc. (BCP), www.neplan.ch/sites/en/neplan_elec.asp

[21] Nexant, Inc., *www.nexant.com/products/PCA/scope1.html*

[22] Electrocon International, Inc., *www.electrocon.com*

[23] EDSA Micro Corporation, *www.edsa.com*

[24] Energy Concepts International, *www.ecieci.com*

[25] Energy Computer Systems, *www.energyco.com*

[26] Powertech Labs Inc., *www.powertechs.com*

[27] Commonwealth Associates, Inc.*, www.cai-engr.com/T2000.htm*

[28] ABB Inc.*, www.abb.com/utilityconsulting*

[29] SKM System Analysis, Inc., *www.skm.com/products/study_modules.htm*

[30] Advantica, *www.stoner.com/stoner_software/synergee_electric/*

[31] MicroTran, *www.microtran.com/index.html*

[32] *IPSA Power Limited,* www.ipsa-power.com/software/software.html

VI. Biographies



Lokendra Bam (Student Member, IEEE) is a graduate student at Wichita State University (WSU) in the Electrical and Computer Engineering Program. He received his Bachelor's degree in Electrical Engineering from Tribhuvan University, Nepal. He worked as consultant engineer for the development and implementation of various renewable energy resources, and rural electric distribution. Currently, he works and assists in research at the Center for Energy Studies at WSU. His research interests include Power System Analysis, Distribution Automation, Distributed Generations, Modeling and Simulation, Power System Communication, Computer Application.



Ward Jewell (M'1977, F'2003) is Director of the WSU Center for Energy Studies and WSU Site Director for the Power Systems Engineering Research Center (pserc.org). Dr. Jewell earned the Ph.D. in Electrical Engineering from Oklahoma State University in 1986. He received the M.S. in Electrical Engineering from Michigan State University in 1980, and the B.S. in Electrical Engineering from Oklahoma State University in 1979.

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on January 25,2024 at 04:11:24 UTC from IEEE Xplore. Restrictions apply.